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South East Technological University

AI PRINCIPLES AND PRACTICE · WEEK 9 OF 12

Adapt, Do Not Rebuild

Transfer learning, foundation models and data augmentation

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Today, in one sentence

Don’t start from zero: reuse a model that already knows a lot, change it a little, and teach it with data, not labels.

REUSE	ADAPT	AUGMENT	COUNT THE COST
don't start from zero	*change a little*	*data, not labels*	*not just accuracy*
pretrained models	**probe · fine-tune · LoRA**	**invariances · synthetic data**	**compute · energy · carbon**

Last week’s maths, this week’s payoff

Week 8 · low rank

A big table can be changed with two thin strips. Today that becomes **LoRA**.

Week 8 · surprise

Big models learn by being less surprised by hidden words or pixels.

Your capstone

Elective E4 is built on transfer learning; every elective can use augmentation.

PART 1

Reuse

Don’t start from zero

YOU ARE HERE

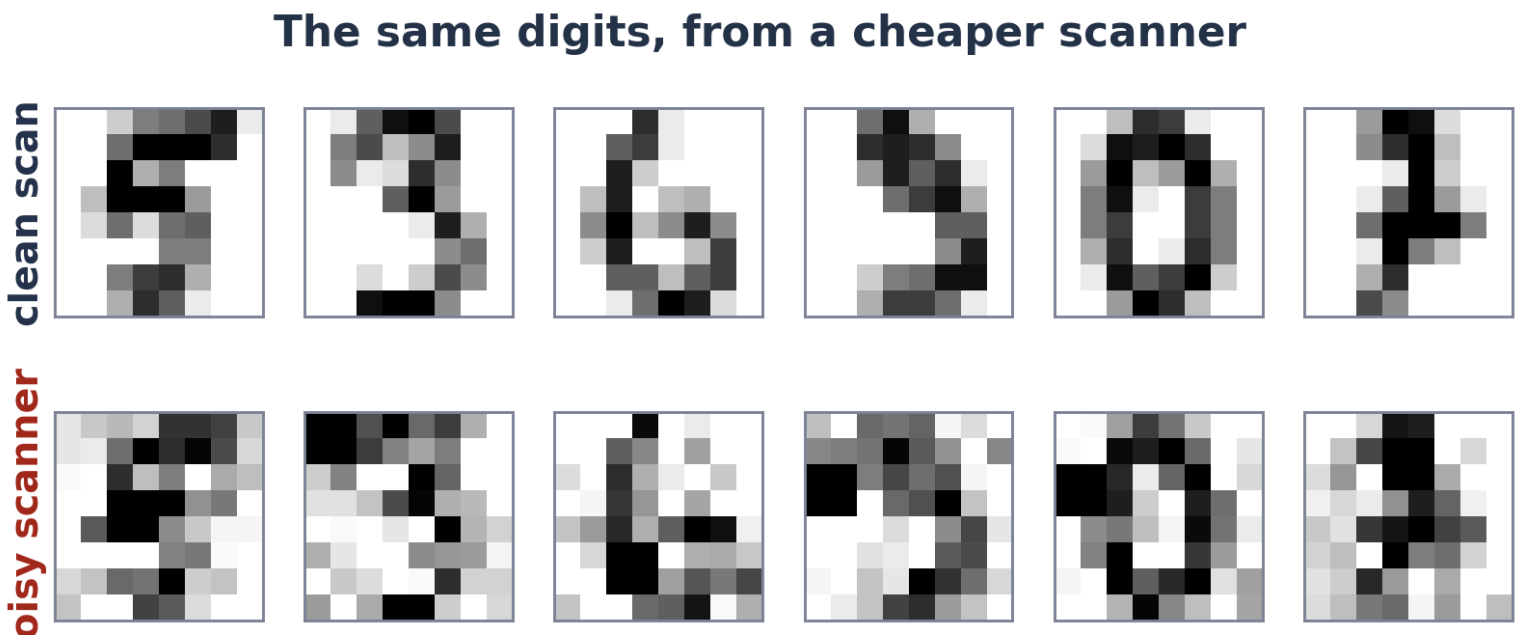
REUSE
don't start from zero

ADAPT
change a little

AUGMENT
data, not labels

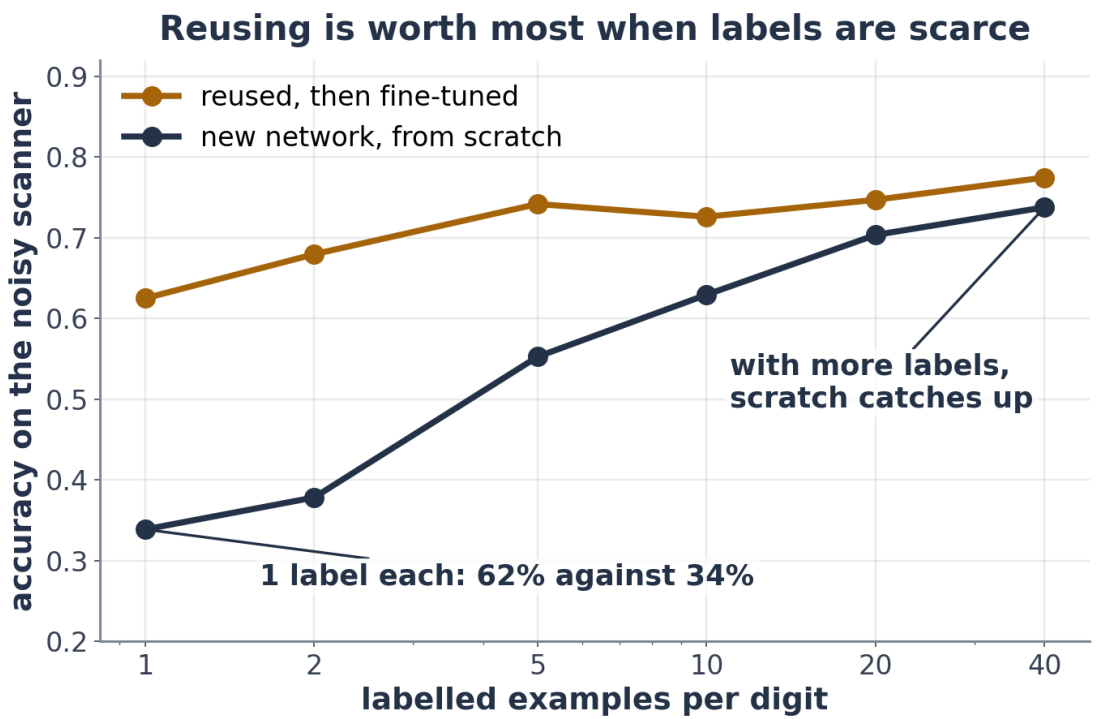
COST
not just accuracy

A new job: digits from a noisy scanner

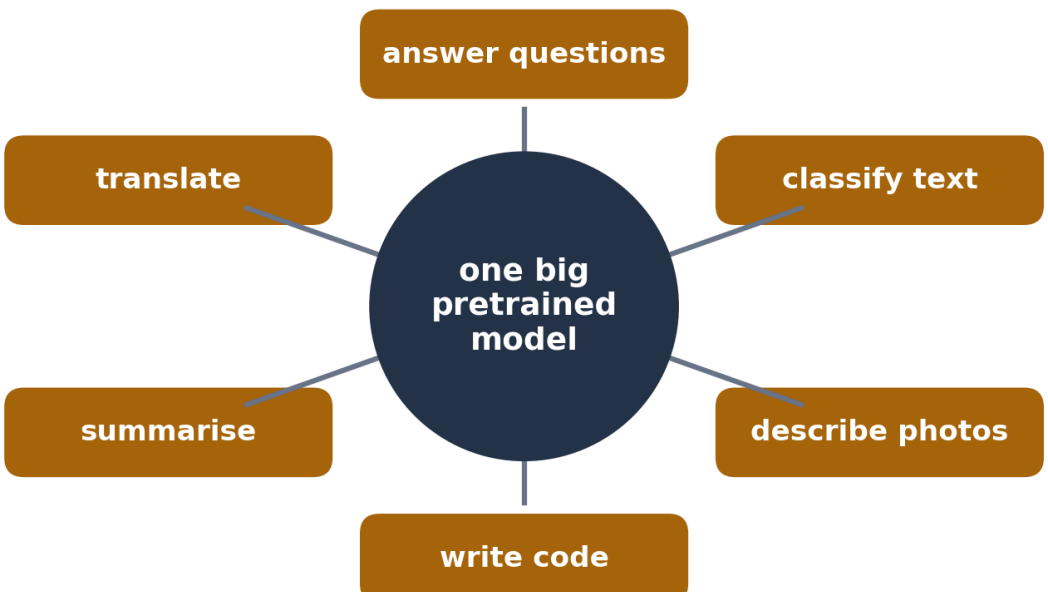


We have a network trained on 900 clean digits, and only 5 labelled examples per digit from the new scanner.

Reusing beats starting again, when labels are few

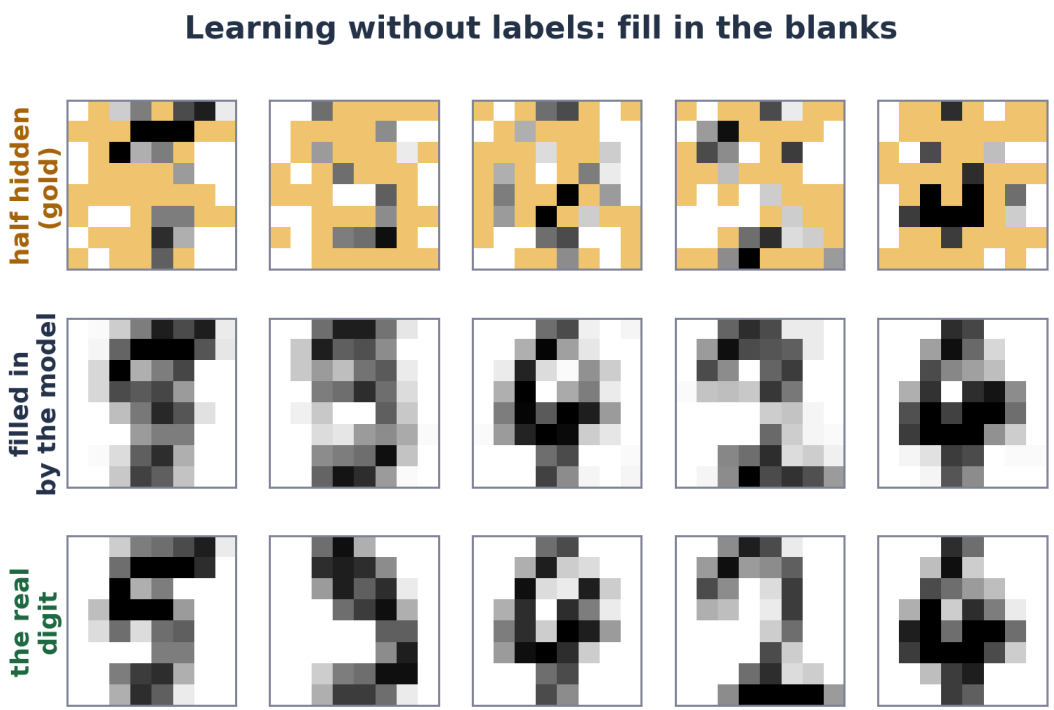


Foundation models: one big model, many jobs



Trained once, on a huge amount of data, at huge cost; then reused for many jobs.

How they learn without labels: fill in the blanks



Fill in the blanks, as a formula

THE FORMULA

$$L = -\,\text{average of}\,\log p(\text{hidden piece} \mid \text{visible pieces})$$

IN WORDS

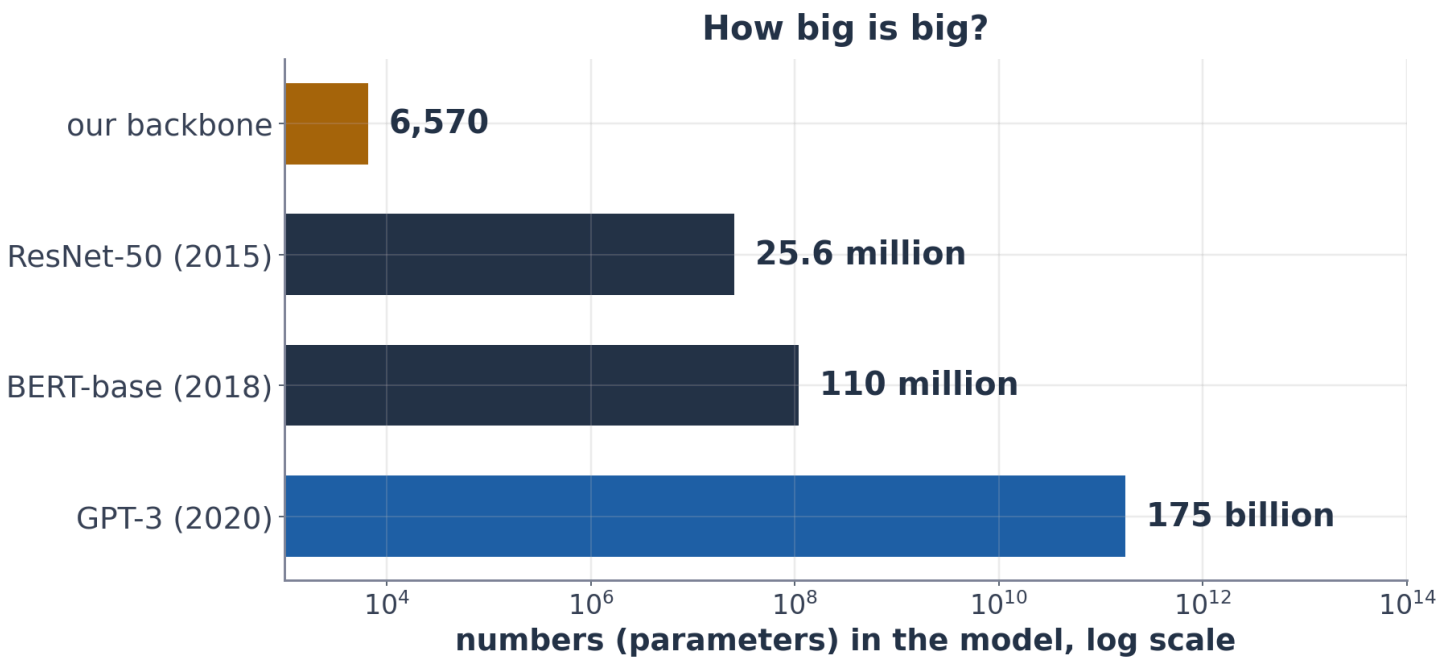
loss = the average surprise at each hidden piece, given the pieces you can see

WITH NUMBERS

“the cat sat on the …”: the model gives “mat” 40%, so the surprise is $-\log_2 0.4 = 1.3$ bits

This is Week 8’s cross-entropy again. The labels come free from the data itself: that is **self-supervised** learning.

How big is big?



Nobody trains these from scratch for one job. Everybody reuses them.

PART 2

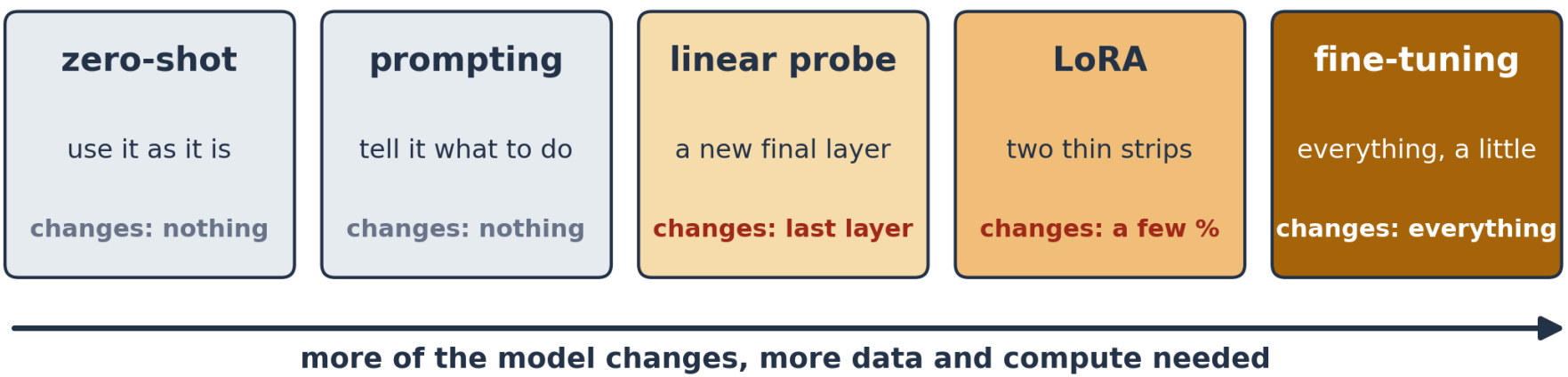
Adapt

Change a little

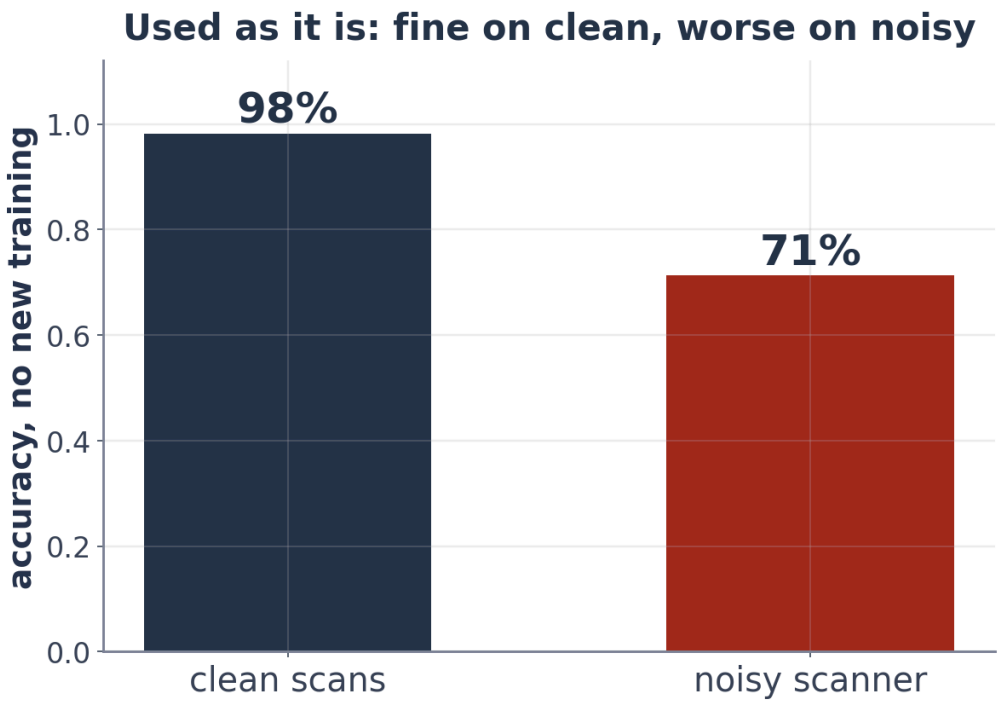
YOU ARE HERE

REUSE	ADAPT	AUGMENT	COST
don't start from zero	change a little	data, not labels	not just accuracy

The transfer spectrum



Use it as it is



Zero-shot: no new training at all. Good when the new job is like the old one.

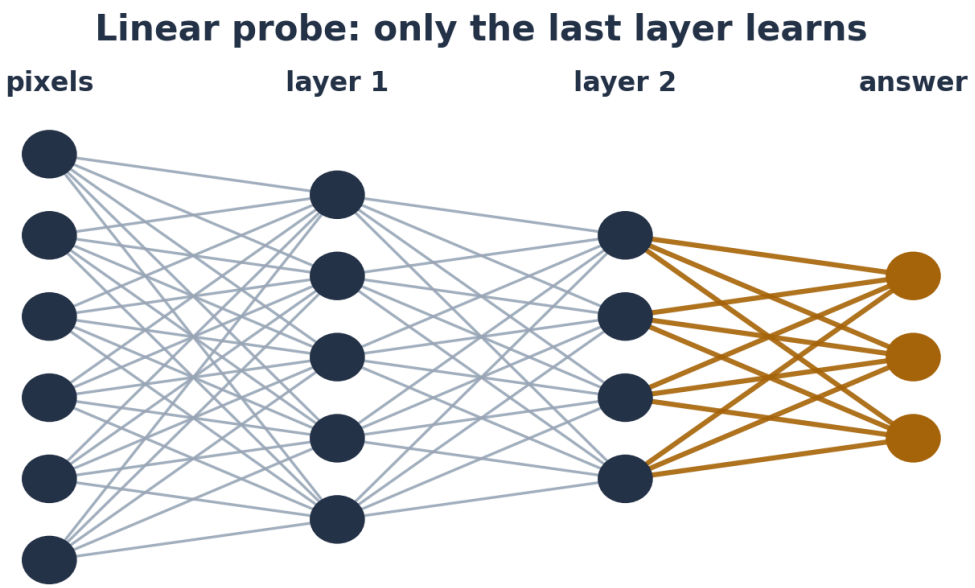
Prompting: change the input, not the model

A PROMPT TO A LANGUAGE MODEL

“Sort each message into **billing**, **delivery** or **other**. Example: ‘My parcel never arrived’ → delivery. Now: ‘I was charged twice’ →”

No number in the model changes. You show it the job, and sometimes a few examples (Week 10).

Linear probe: a new head on a frozen body



Linear probe, as a formula

THE FORMULA

$$\hat{y} = \mathrm{softmax}(W_{\mathrm{head}} \cdot f(x))$$

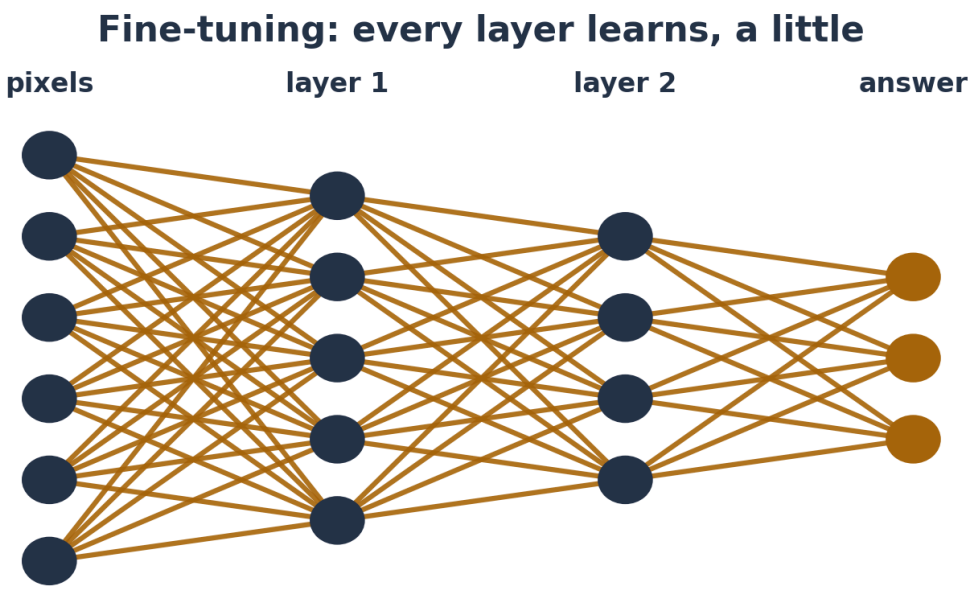
IN WORDS

answer = a new last layer applied to what the frozen network sees; only that last layer learns

WITH NUMBERS

330 numbers trained: 71% on the noisy scanner

Fine-tuning: every layer learns, a little



Fine-tuning, as a formula

THE FORMULA

$$\theta \leftarrow \theta - \eta \nabla L, \text{starting from } \theta_{\text{pretrained}}$$

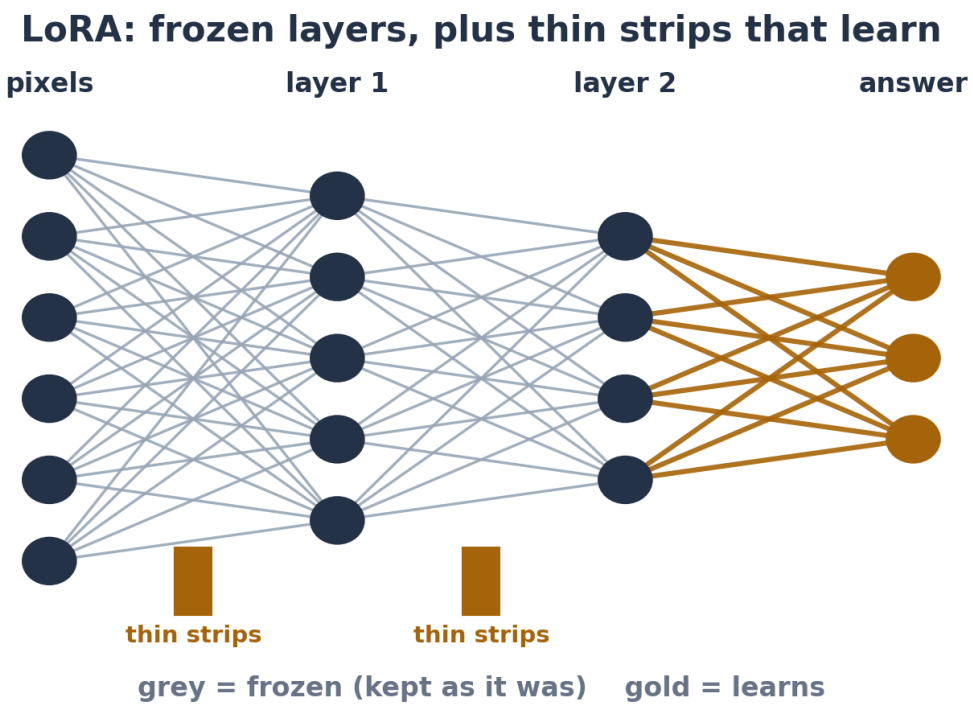
IN WORDS

the same downhill step as Week 7, but starting from the pretrained dials, with a small step

WITH NUMBERS

all 6,570 numbers trained: 74% on the noisy scanner

LoRA: frozen layers, plus thin strips



LoRA, as a formula

THE FORMULA

$$W' = W + BA$$

IN WORDS

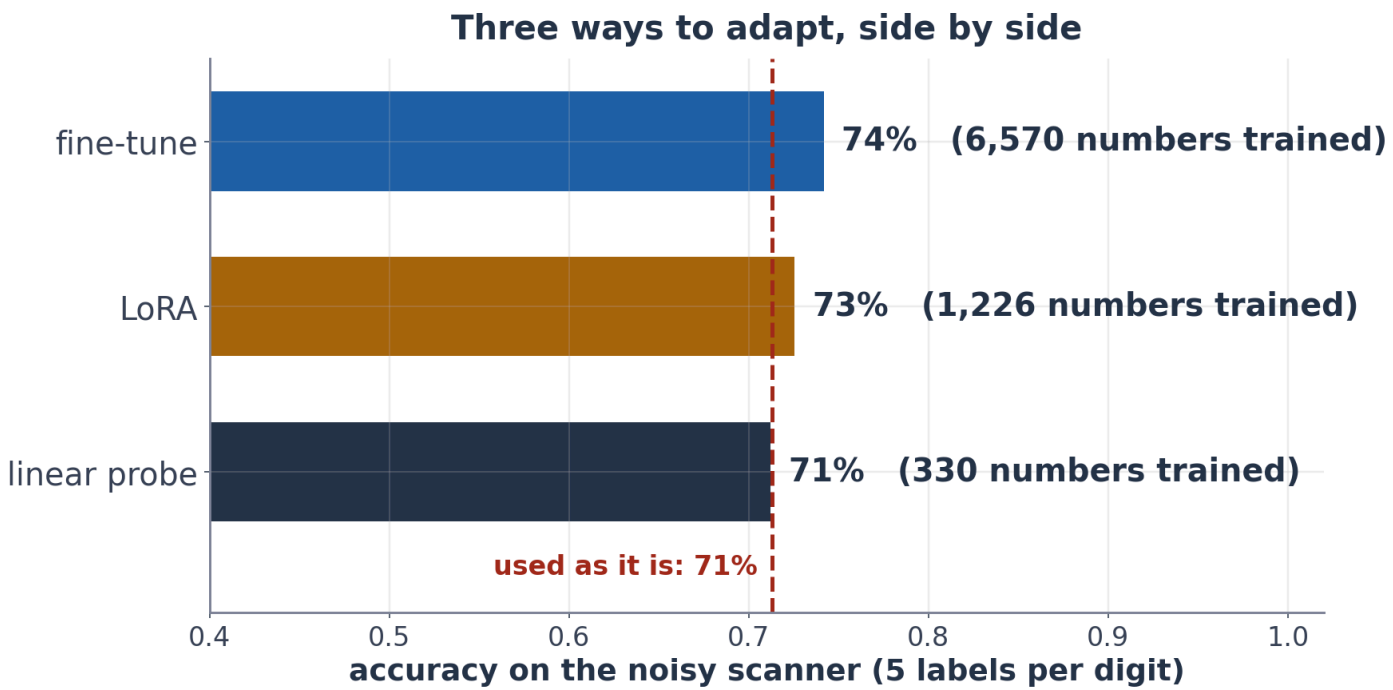
new weights = old weights (frozen) + thin strip × thin strip

WITH NUMBERS

rank 4 here: 1,226 numbers trained, 73% · a 768 × 768 table at rank 8: 12,288 instead of 589,824

For GPT-3, LoRA cut the numbers to train by about 10,000 times (Hu and others, 2021).

Three ways, side by side



Which should you choose?

Use it as it is	the new job is like the old one; you have no labels
Prompt	a language model; a handful of examples fit in the prompt
Linear probe	few labels, little compute; a quick, honest baseline
LoRA	a big model, a real change needed, limited memory or budget
Fine-tune	plenty of labels and compute, and the new job is very different

PART 3

Augment

Data, not labels

REUSE	ADAPT	AUGMENT	COST
don't start from zero	change a little	data, not labels	not just accuracy

YOU ARE HERE

More examples, without more labels



Copies that change the picture but not the answer, and one that changes the answer.

Augmentation says what does not matter

EVERYDAY ANALOGY

A smudged 6 is still a 6. A 6 a little to the left is still a 6. By showing the model smudged and shifted copies with the same label, you tell it: **ignore smudges and shifts**. That is called an **invariance**.

Choosing the copies is choosing what the model learns to ignore.

Invariance, as a formula

THE FORMULA

$$f(T(x)) = f(x)$$

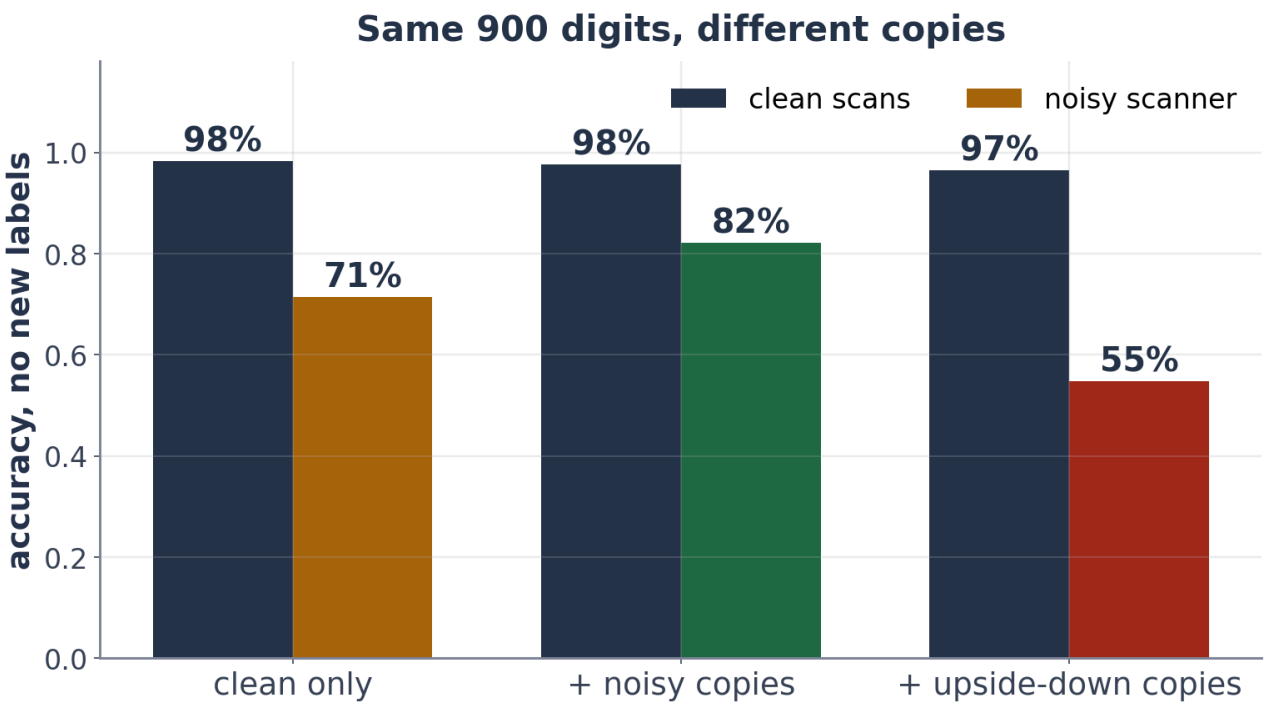
IN WORDS

the answer should not change when you apply the change T

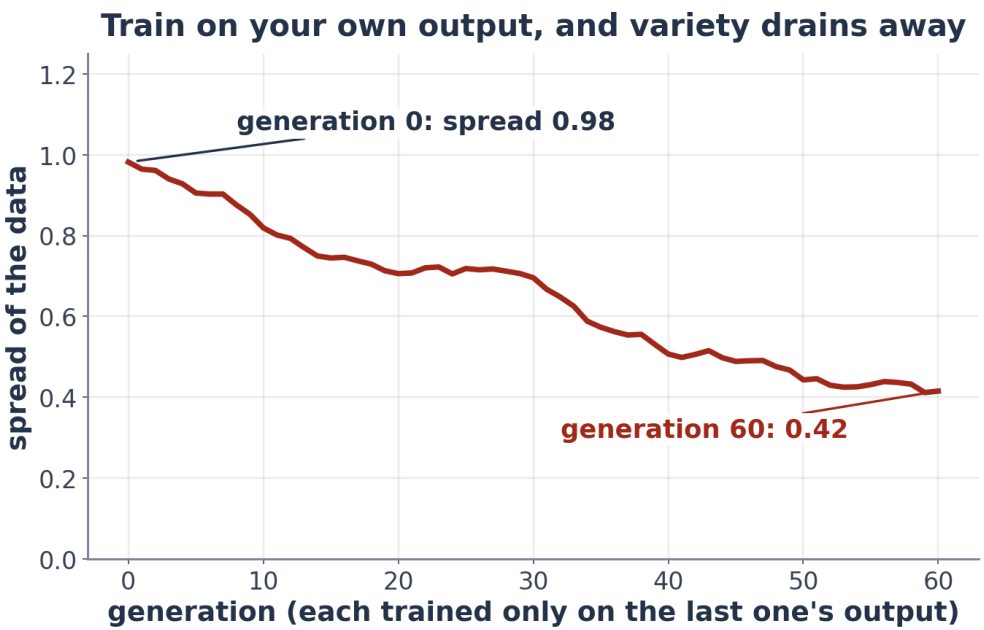
WITH NUMBERS

T = add scanner noise: 71% → 82% on the noisy scanner, with no new labels

The right copies help; the wrong ones hurt



Synthetic data: useful, with care



Data made by a model can fill gaps; a model fed only on its own output slowly loses variety.

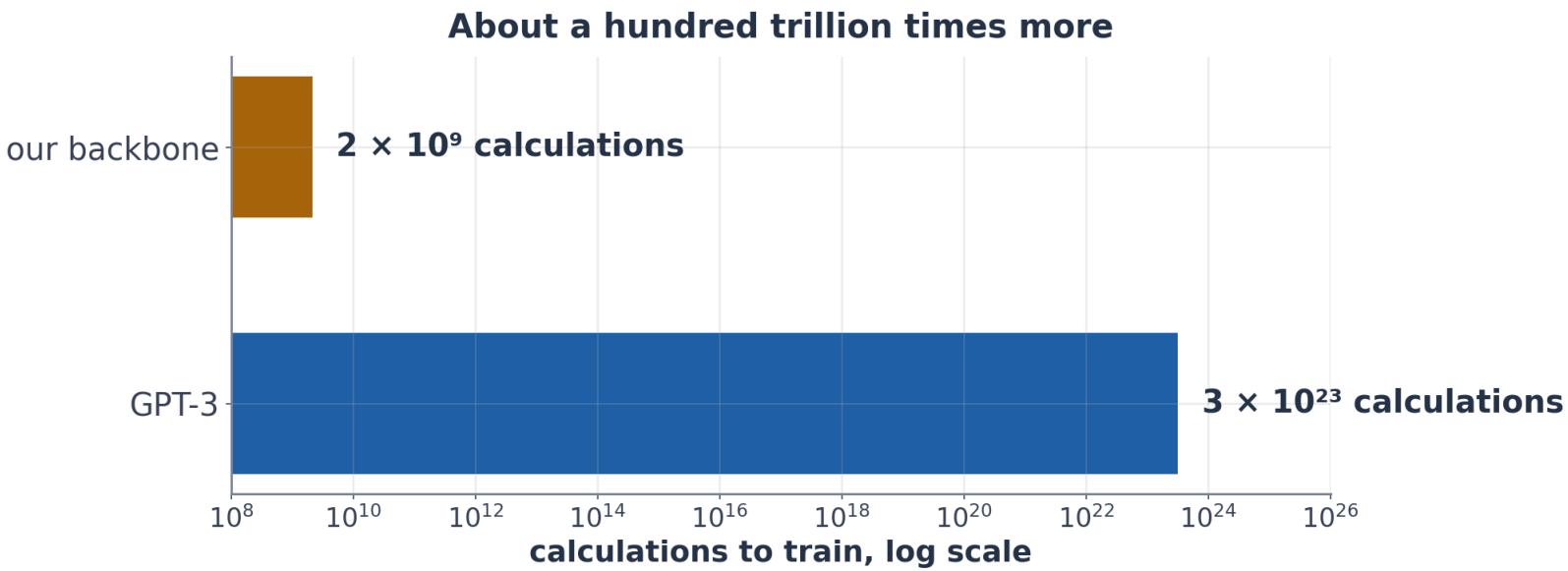
PART 4

Count the cost

Not just accuracy

REUSE	ADAPT	AUGMENT	YOU ARE HERE COST
don't start from zero	change a little	data, not labels	not just accuracy

Bigger is not free



Compute, as a formula

THE FORMULA

$$C \approx 6ND$$

IN WORDS

calculations to train ≈ 6 × numbers in the model × examples it learns from

WITH NUMBERS

GPT-3: 6 × 175 billion × 300 billion words ≈ 3×10^{23}

Energy and carbon, as formulas

THE FORMULA

$$E = Pt, \quad \mathrm{CO}_2 = E \times I$$

IN WORDS

energy = power × time; carbon = energy × the grid’s carbon per kWh

WITH NUMBERS

300 W for 10 h = 3 kWh ≈ 0.9 kg CO_2 · GPT-3: about 1,287 MWh and 552 t CO_2

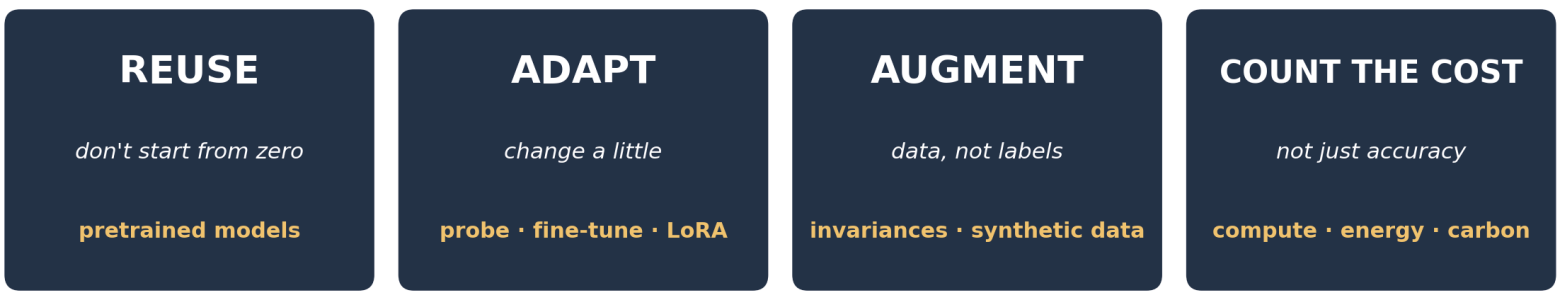
GPT-3 figures from Patterson and others (2021). 0.3 kg per kWh is an example grid; it varies by country and hour.

Choose by more than accuracy

Way	Accuracy	Numbers trained	Memory, time, carbon
Linear probe	71%	330	least
LoRA	73%	1,226	a little
Fine-tune	74%	6,570	most; and a full copy to store

On a big model, the gap in cost is millions of times larger than here; the gap in accuracy often is not.

Reuse, adapt, augment, count the cost



- **Reuse:** with few labels, a pretrained model beats starting from scratch.
- **Adapt:** as it is, prompt, probe, LoRA or fine-tune: change only as much as the job needs.
- **Augment:** copies that keep the answer teach the model what to ignore, without new labels.
- **Count the cost:** compute, energy and carbon are part of choosing a model.

Where this fits in your elective

E1, E2	Tabular data: a linear probe on learned features is a strong baseline; augment only with changes that keep the meaning.
E3	A pretrained language model: prompt it, probe it, or fine-tune it with LoRA.
E4	Built on this week: pretrained image and text models, adapted three ways, with their costs side by side.
E5, E6	No pretrained model, but the same idea: build what you know (the physics) into the model instead of learning it.

Where these ideas come back

Week 10 Transformers: the models inside most foundation models, and why prompting works.

Week 11 Diffusion models: generating new data, and the care synthetic data needs.

Adapt, Do Not Rebuild

1. Reuse the backbone as it is, then against a network trained from scratch.
2. Adapt it three ways: linear probe, LoRA and fine-tuning; compare accuracy, numbers trained and time.
3. Improve it with augmentation alone, and see the wrong augmentation hurt.
4. Count the cost: compute, energy and carbon.

Everything runs in NumPy in a few seconds. Not graded this week.

REUSE
don't start from zero

ADAPT
change a little

AUGMENT
data, not labels

COST
not just accuracy

Reuse. Adapt. Augment. Count the cost.

Questions?